

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**  
**Bachelor of Technology (Computer Science and Engineering) SEMESTER - 4 Summer 2026 (Regular)**  
**Course : Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology**  
**Semester : SEMESTER - 4**  
**Subject Code & Name: 25AF1245PC401 - DESIGN AND ANALYSIS OF ALGORITHMS**

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

**Q1. Objective type questions. (Compulsory Question)**

**12**

- 1 A recursive algorithm gives you a recurrence relation. Which approach would you apply to estimate the total cost of recursive calls?  
A. Change variables  
B. Use a recursion tree  
C. Convert to an iterative loop  
D. Write pseudocode

- 2 Which sequence of operations leads to an incorrect max-heap during heap sort?  
A. Heapify after swapping root with last  
B. Building max-heap starting from the last non-leaf node  
C. Swapping the root with any middle element before heapifying  
D. Reducing heap size by 1 after extracting max

- 3 

```
Void Test (int n)
{
    if(n>0)
    {
        for (i=0; i<n; i++)
        {
            Printf ("%d", n);
        }
        Test(n-1)
    }
}
```

Find the correct recurrence relation for given algorithm

- A.  $T(n) = T(n-1) + n$   
B.  $T(n) = T(n-1) + (n+1)$   
C.  $T(n) = T(n) + n$   
D.  $T(n) = T(n+1) + n$
- 4 Why is Binary Search more efficient than Linear Search for large sorted datasets?  
A. It checks all elements simultaneously  
B. It reduces the search space by half in each step  
C. It sorts the data during searching  
D. It uses recursion only

- 5 Strassen's Matrix Multiplication algorithm is preferred over the conventional method because it:
- Requires less memory only
  - Uses fewer multiplication operations
  - Eliminates addition operations
  - Works only for square matrices
- 6 Which statement best describes the outcome of the partitioning phase in Quick Sort?
- The pivot is correctly positioned, creating two subarrays for further sorting
  - The array is fully sorted in a single pass
  - The largest element is moved to the end of the array
  - Adjacent elements are repeatedly swapped
- 7 Which statement about Branch and Bound is correct?
- It never explores partial solutions
  - It uses bounds to avoid exploring unpromising branches
  - It is used only for sorting problems
  - It guarantees constant-time execution
- 8 Which principle of Huffman Coding helps reduce the overall size of encoded data?
- Sorting symbols alphabetically
  - Assigning equal-length codes to all symbols
  - Assigning shorter codes to frequently occurring symbols
  - Removing repeated symbols
- 9 Why is the Optimal Merge Pattern considered a greedy strategy?
- It always merges the two smallest files first to reduce total merging cost
  - It sorts files before merging them
  - It merges files in random order
  - It minimizes memory usage only
- 10 Which of the following best describes the Sum of Subsets problem?
- Finding a group of elements that collectively achieve a desired sum
  - Finding the largest element in a set
  - Partitioning a set into equal subsets
  - Sorting subsets based on their sums
- 11 Why is Dynamic Programming often more efficient than Divide-and-Conquer for certain problems?
- It reduces the input size more quickly
  - It avoids repeated computation of overlapping subproblems through storage of intermediate results
  - It eliminates the need for recursion
  - It uses less memory
- 12 Which of the following best describes the complexity class P?
- Their solutions are difficult to verify
  - Problems that can be solved in polynomial time by a deterministic algorithm
  - Problems that require exponential time to solve
  - Problems that cannot be solved efficiently

Q2. Solve the following.

- A) Explain the concepts of Time Complexity and Space Complexity with a suitable example. 6

- B) Construct a max heap to sort the elements in descending order, and show all intermediate steps: 45, 20, 15, 30, 10, 50, 25 6

Q3. Solve the following.

- A) Explain working of merge sort with suitable example. Also write algorithm for merging of element and recursive merge sort. 6
- B) Justify why Strassen's Matrix Multiplication algorithm has a lower time complexity than the conventional matrix multiplication algorithm. 6

Q4. Solve Any Two of the following.

- A) Using Backtracking, determine whether the set {5, 10, 12, 13, 15, 18} contains a subset whose sum is 30. Draw the state space tree. 6
- B) Explain the concept of Graph Coloring. How are colors assigned to vertices while satisfying coloring constraints? Illustrate with an example. 6
- C) Compare Backtracking and Branch and Bound techniques with respect to their search strategies and applications. 6

Q5. Solve Any Two of the following.

- A) A file contains the following characters with the frequencies as shown. If Huffman coding is used for data compression determine  
1. Huffman code for each character 2. Length of Huffman code encoded message. 6

| Characters | frequencies |
|------------|-------------|
| P          | 1           |
| Q          | 5           |
| R          | 4           |
| S          | 2           |
| T          | 2           |
| U          | 1           |
| V          | 1           |
| W          | 1           |

- B) Solve the following fractional knapsack problem and check the weight constraint for given knapsack and profit gain. The given data - <https://www.batuonline.com>  
 $p = (p_1, p_2, p_3, p_4, p_5, p_6) = (18, 5, 9, 10, 12, 7)$   
 $w = (w_1, w_2, w_3, w_4, w_5, w_6) = (7, 2, 3, 5, 3, 2)$   
 $M = \text{Knapsack Capacity} = 13$ . 6
- C) Explain how Prim's algorithm differs from Kruskal's algorithm in building a Minimum Spanning Tree. 6

Q6. Solve Any Two of the following.

- A) Explain the complexity classes P, NP, and NP-Complete with suitable examples. 6
- B) Solve the following matrix chain multiplication using dynamic programming  
A (4x5) B (5x3) C (3x10) D (10x2) 6
- C) Explain the Longest Common Subsequence (LCS) problem. Using Dynamic Programming, find the LCS of the strings X='ABCBADAB' and Y='BDCAB'. 6

\*\*\* End \*\*\*